



SUJET STATAPP 298 - 2026-2027



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Organisme : ENGIE

Catégorie : Prédiction

Sujet :

Belgium's Hourly Gas Consumption Model: Accuracy and Relevance

Description du sujet :

Objective: Assess the accuracy of Engie's predictive model for hourly natural gas consumption for its Belgian clients, to support the dispatch department in ensuring efficient and reliable gas distribution.

Approach:

? Engie's predictive model forecasts hourly natural gas consumption for its Belgian clients (GAM+Residuals adjustment)

? It typically forecasts three days ahead.

? Explanatory variables are climate- and temperature-dependent

Key Issue:

? The model was developed some time ago, and it is important to regularly update and recalibrate it ? or even completely change it if a better approach is revealed.

Consequence / Mismatch:

? Inaccuracy can result in inefficient supply strategies, causing either surpluses or shortages and increasing costs.

? Moreover, an outdated model might fail to identify new risks, such as shifting demand patterns, leaving the dispatch department unaware of those changes.

Task:

Ensure the model remains accurate and relevant by evaluating gradient tree-based approaches, as well as combinations with existing models.

Données du sujet :

hourly consumptions, temperatures, wind, calendar

Modèles envisagés du sujet :

tree-based models

Bibliographie du sujet :

look for gradient boosting models on ScienceDirect

Notation du sujet :

Commentaires :